Clinical Note Specialty Classification — Model Report

- Repo root: /Users/aidancolvin/health-data-nlp-api
- Processed data dir: /Users/aidancolvin/health-data-nlp-api/data/processed
- Data split: train/val/test created by src.utils.prepare_data with train_size=0.80, val_size=0.10, test_size=0.10.
- Classical models trained: logistic regression, SVM, random forest, gradient boosting.
- Classical CV: 5-fold cross-validation (src.model.classical.cv_classical), metrics saved to data/processed/metrics_cv_*.csv.
- Test eval: src.model.compare_models outputs f1_macro and accuracy and writes data/processed/leaderboard.csv (if enabled).
- Transformer: DistilBERT fine-tuning uses Transformers v5 API (eval_strategy; Trainer processing_class).

Table 1. Classical CV summary (5-fold)

model	accuracy_mean	accuracy_std	f1_macro_mean	f1_macro_std	f1_weighted_mean	f1_weighted_std
svm	0.154332	0.00380289	0.120294	0.0165719	0.148308	0.00441361
rf	0.146776	0.00933541	0.0654723	0.00447072	0.134508	0.00937949
logreg	0.274422	0.0133323	0.0631905	0.0122841	0.212751	0.0128847
gb	0.151032	0.0540323	0.0404175	0.0221915	0.106559	0.0300555

Table 2. Combined leaderboard (what is currently available)

model	split	accuracy	f1_macro	source
svm	cv5_mean	0.154332	0.120294	classical_cv
rf	cv5_mean	0.146776	0.0654723	classical_cv
logreg	cv5_mean	0.274422	0.0631905	classical_cv
gb	cv5_mean	0.151032	0.0404175	classical_cv
logreg	test	nan	nan	classical_test
svm	test	nan	nan	classical_test
rf	test	nan	nan	classical_test
gb	test	nan	nan	classical_test
distilbert	test	nan	nan	classical_test

Figure 1. Classical models: CV macro-F1 (mean).

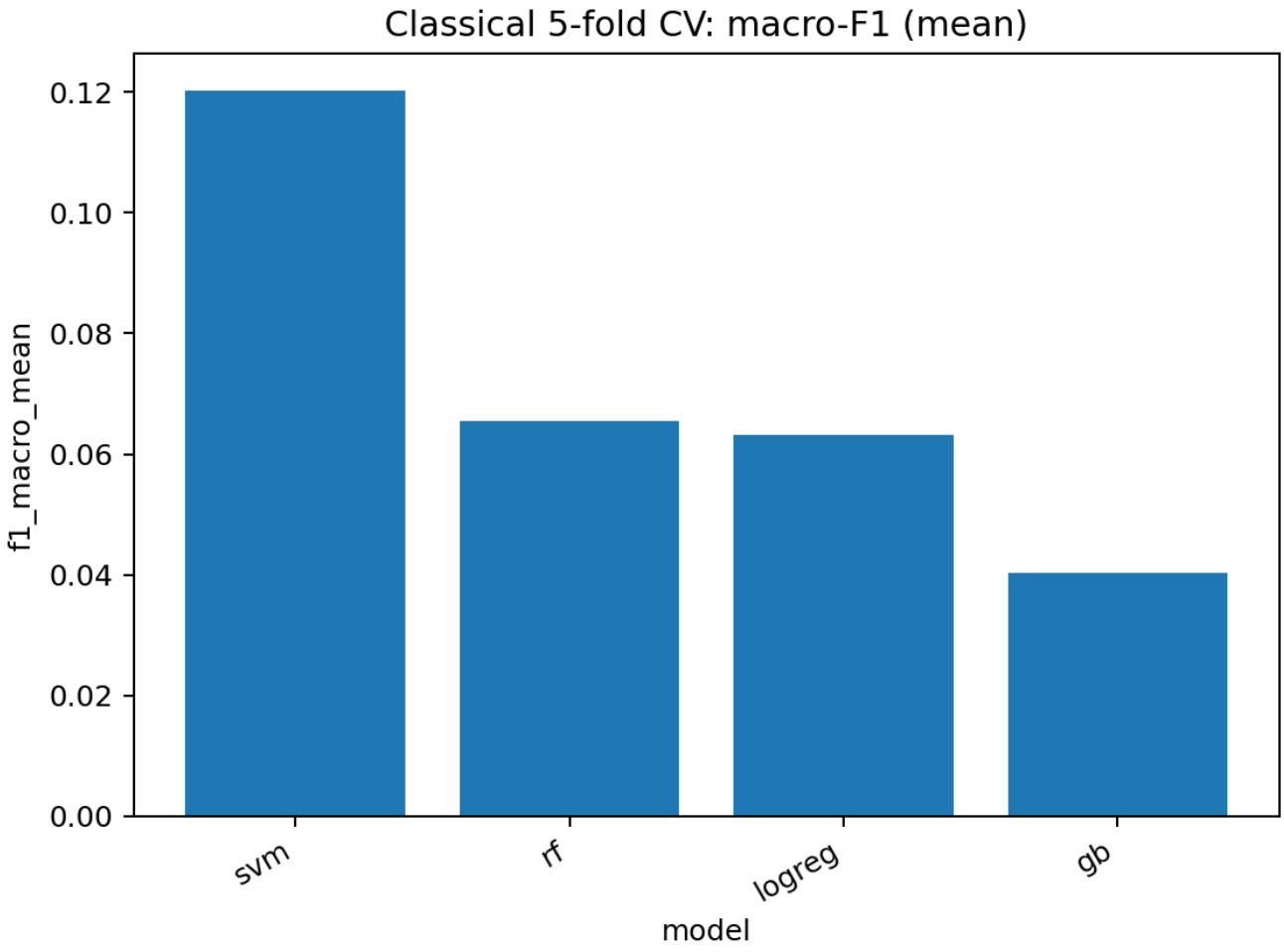


Figure 2. Classical models: CV accuracy (mean).

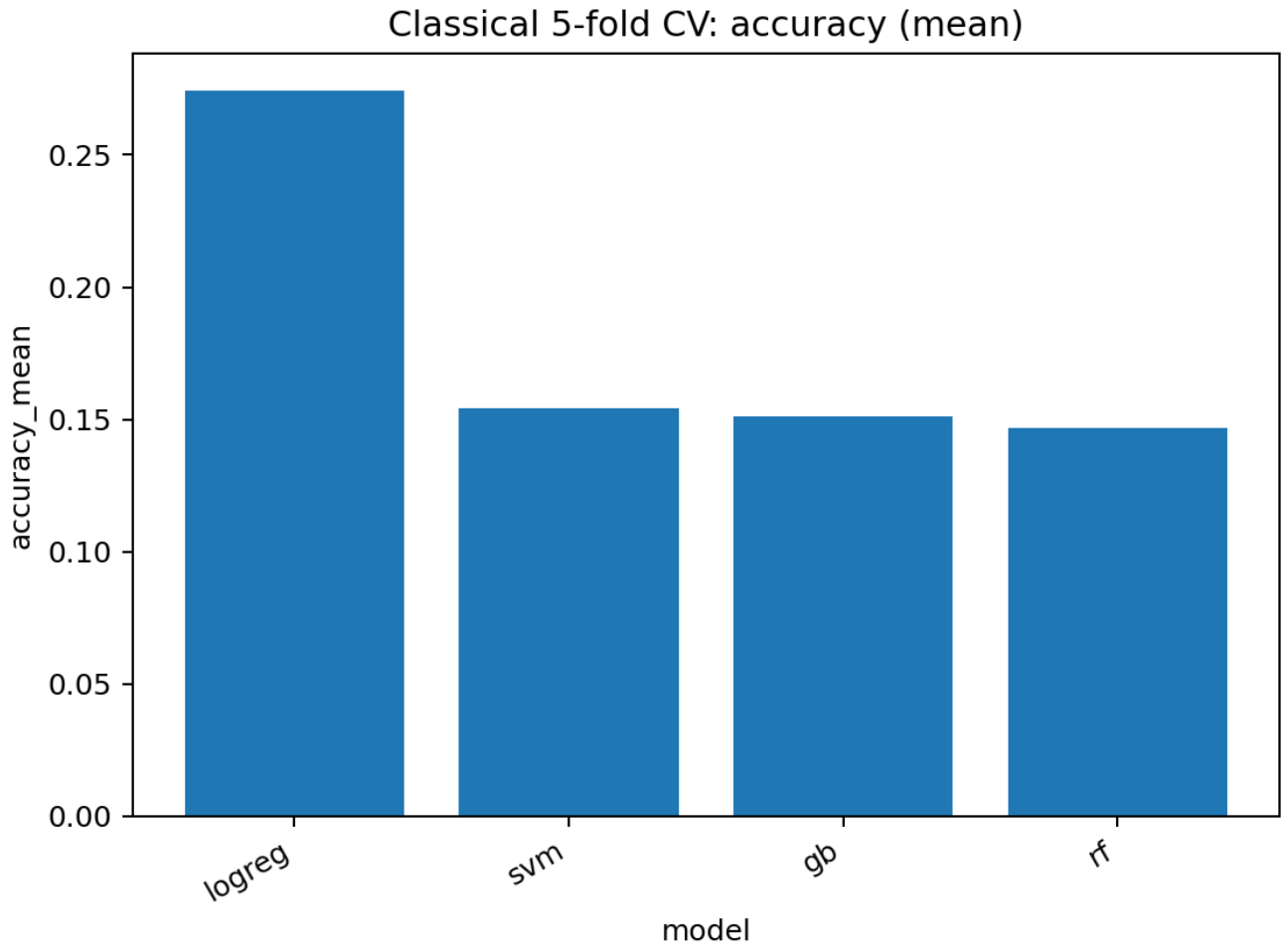


Figure 3. Macro-F1 available so far (CV + test where present).

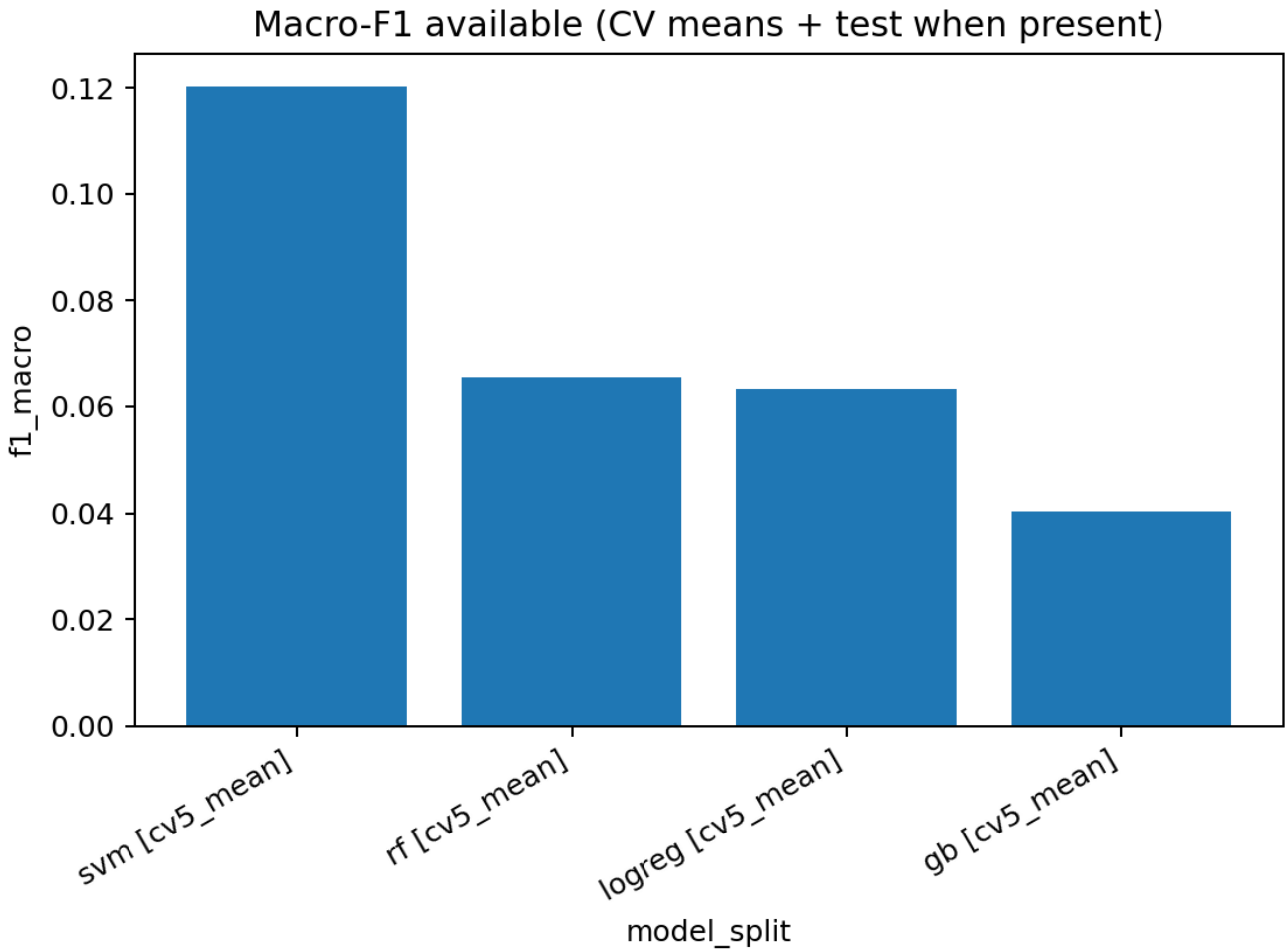
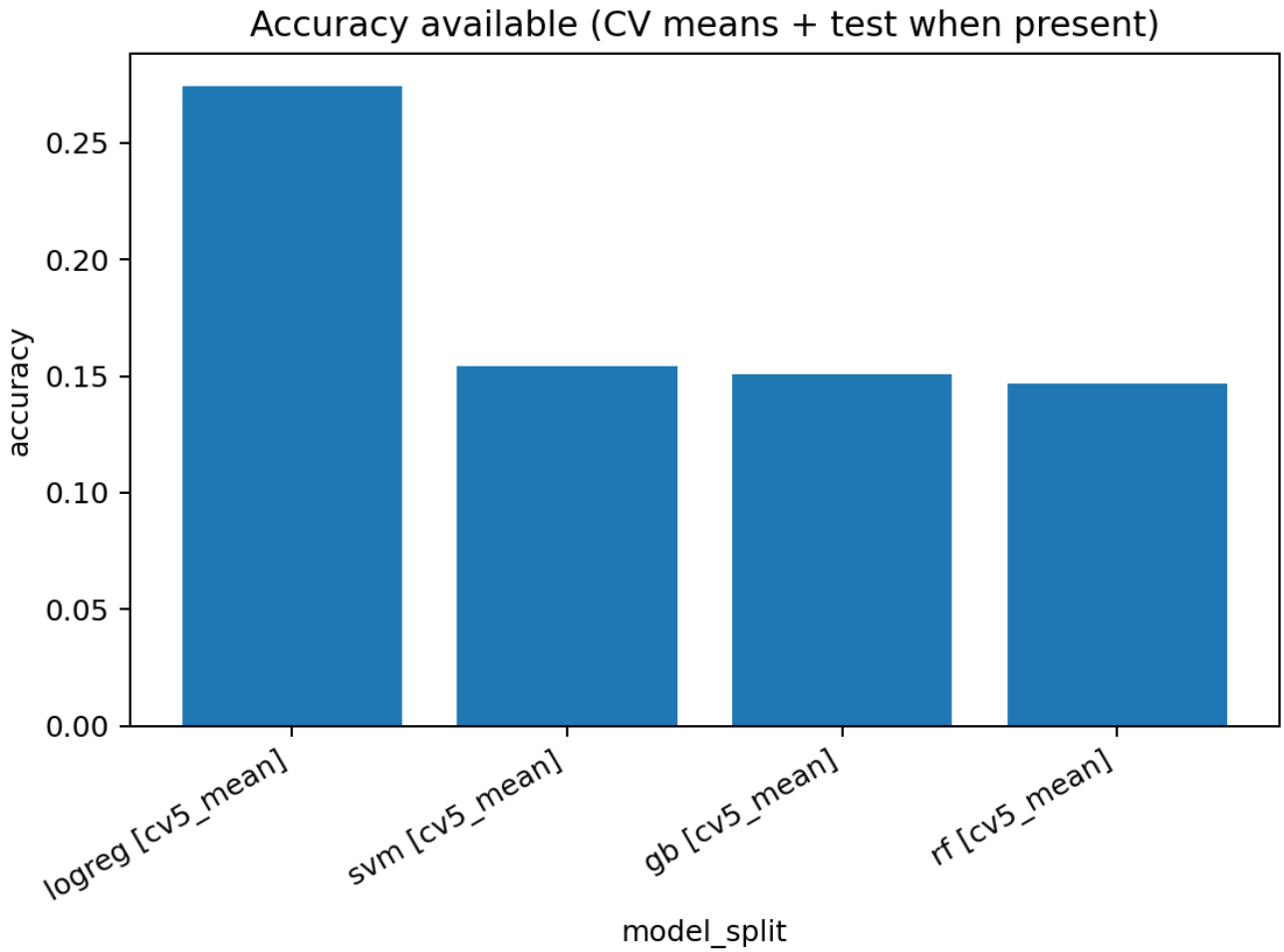


Figure 4. Accuracy available so far (CV + test where present).


Accuracy available (CV means + test when present)
accuracy
0.25
0.20
0.15
0.10
0.05
0.00
logreg [cv5_mean]
svm [cv5_mean]
gb [cv5_mean]
rf [cv5_mean]
model_split

Results Notes (based on saved artifacts)

- Best CV macro-F1: svm = 0.120294 (± 0.0165719).
- Best CV accuracy: logreg.
- DistilBERT: local model artifacts found at /Users/aidancolvin/health-data-nlp-api/models/transformer/distilbert.

Limitations (tight)

- Class imbalance is severe (long-tail specialties). Macro-F1 stays low even when accuracy rises.
- Your combined table shows NaN test metrics for several rows. Those scripts must write numeric fields to data/processed/combined_leaderboard.csv.
- Transformer metrics must be produced by a working evaluate module that writes results back into the same combined artifact.